

DRAFT

FA8609-23-F-0008
Section J, Attachment 1
4 Mar 2026

**STATEMENT OF WORK (SOW)
FOR**

**F139 (PW 4062-3) ZERO-TIME SPARE ENGINES (CORE ONLY)
FOR
USAF REQUIREMENTS
Delivery Order Contract No.: TBD**

4 Mar 2026

PREPARED BY

**AFLCMC/LPS
PROPULSION ACQUISITION DIVISION
TINKER AIR FORCE BASE, OKLAHOMA 73145**

On Behalf of

**AFLCMC/WLC
MOBILITY AND TRAINING AIRCRAFT DIRECTORATE
KC-46 DIVISION
WRIGHT-PATTERSON AIR FORCE BASE, OHIO 45433**

Controlled by: USAF

Controlled by: AFLCMC/LPS

CUI Category: EXPT, PROCURE

Distribution Statement D

POC: AFLCMC.WKCWorkflow@us.af.mil

DISTRIBUTION STATEMENT D - Distribution authorized to the Department of Defense (DoD) and U.S. DoD contractors only Export Controlled, XX XXX 2026. Other requests for this document shall be referred to the F139 Logistics Propulsion Sustainment Division, 3001 Staff Dr, TAFB, OK 73145.

WARNING – This document contains technical data whose export is restricted by the Arms Export Control Act (Section 2751 of Title 22, United States Code) or the Export Administration Act of 2018, -(Chapter 58 Sections 4801-4852 of Title 50, United States Code). Violations of these export laws are subject to severe criminal penalties. Disseminate in accordance with provision of DoD Directive 5230.25 and DoD Instruction 2040.02.

CUI

DRAFT

CUI

VERSION CONTROL

VERSION	DATE	COMMENT

Table of Contents

1.0 INTRODUCTION AND SCOPE	5
2.0 APPLICABLE DOCUMENTS	5
3.0 MODIFICATIONS, ENHANCEMENTS, and UPGRADE SUPPORT - F139 SPARE ENGINES	6
3.1 PW4062-3 Engine.....	6
3.2 Manufacturing Management Program.....	6
3.2.1 Bill of Material (BOM)	6
3.3 Contractor Furnished Fuel	7
3.4 Logistics.....	7
3.4.1 Comprehensive Engine Management System (CEMS).....	7
3.4.2 Engine Item Unique Identification (IUID) Markings.....	7
3.4.3 Packaging, Handling, Storage, and Transportation (PHS&T)	8
3.4.4 Shipping of Spare Engines.....	8
4.0 TEST AND EVALUATION - PRODUCT ACCEPTANCE.....	8
4.1 Acceptance Test Plans and Acceptance Test Procedures (ATPs)	8
4.2 Acceptance Test Records.....	Error! Bookmark not defined.
5.0 SYSTEMS ENGINEERING/PROJECT MANAGEMENT (SEPM)	8
6.0 PROGRAM MANAGEMENT	8
6.1 Contract Work Breakdown Structure.....	8
6.2 Integrated Master Schedule (IMS)	9
6.3 Cost and Software Data Reporting.....	9
6.4 Risk Management.....	9
6.5 Government Industry Data Exchange Program (GIDEP)	9
7.0 CONFIGURATION MANAGEMENT.....	10
7.1 Contract Change Proposal (CCP)	10
7.2 Specification Change Notice (SCN).....	10
7.2.1 Specification Maintenance	Error! Bookmark not defined.
7.3 Deviations and Waivers	10
8.0 DATA MANAGEMENT.....	10
9.0 RESERVED	10
10.0 RESERVED	10
11.0 RESERVED	10
12.0 RESERVED	10

13.0 QUALITY ASSURANCE	10
13.1 Foreign Object Damage (FOD) Prevention Program	11
14.0 CONTINUED AIRWORTHINESS AND REGULATORY COMPLIANCE	11
15.0 SYSTEM SAFETY	11
16.0 ENVIRONMENTAL FACTORS/ENVIRONMENT SAFETY AND OCCUPATIONAL HEALTH (ESOH)	11
17.0 SECURITY	11
18.0 KC-46 INFORMATION TECHNOLOGY	11
19.0 RESERVED	11

1.0 INTRODUCTION AND SCOPE

The KC-46A aircraft is equipped with two Pratt & Whitney (P&W) PW4062-3 commercial engines . The KC-46A engine military equivalent designation is the F139 engine. General capabilities/specifications of the KC-46A F139 engine include thrust per high-bypass engine of 62,000 lbs.; maximum speed of Mach 0.86; cruise speed of Mach 0.80; range of 6,385 NM; and service ceiling of 40,100 ft. The F139 is a modified commercial engine which varies from the PW4062-3 in that it does not have thrust reversers installed. However, the F139 does have an increased thrust capability termed “thrust bump,” additional electro-magnetic/IR shielding, and a larger generator as compared to the PW4062-3.

This SOW defines the effort required to meet the USAF requirements for the production and delivery of 12 spare engine cores for the KC-46A fleet, with the procurement option of seven additional engines.

The USAF will procure F139 spare engine cores in the most current configuration at the time of award.

Each spare engine shall be preserved, wrapped, and shipped to Tinker AFB where Engine Build Up (EBU) will occur making the spare engine Ready for Installation. Also included in the requirements is the delivery of complete engine build data for each spare engine core. The Contractor shall deliver all engine build data to AFLCMC/LPSD Program Management Office and Engineering Office.

2.0 APPLICABLE DOCUMENTS

The below listed reference documents form a part of this procurement to the extent specified herein.

USG Documents		
Document Number	Document Name	Version and Date
DoDM 5000.04	<i>Cost and Software Data Reporting (CSDR) Manual</i>	7 May 2021
MIL-DTL-85470C	<i>Inhibitor, Icing, Fuel System, High Flash NATO Code Number S-1745</i>	5 Mar 2024
MIL-PRF-25017H	<i>Inhibitor, Corrosion/Lubricity Improver, Fuel Soluble (NATO S-1747)</i>	25 May 2021
MIL-STD-129R	<i>Military Marking for Shipment and Storage</i>	25 Feb 2023
MIL-STD-130N	<i>Identification Marking of U.S. Military Property</i>	16 Nov 2012
MIL-STD-881F	<i>Work Breakdown Structures for Defense Material Items</i>	13 May 2022
MIL-STD-2073-1E	<i>Department of Defense Standard Practice for Military Packaging</i>	27 Jun 2024
AFMC Form 158	<i>Packaging Requirements</i>	14 Oct 2022
AFMC Instruction 24-201	<i>Packaging, Handling, Storage, and Transportation Acquisition and Sustainment Product Support Instructions</i>	21 June 2022
Technical Order (TO) 00-25-254-1	<i>Comprehensive Engine Management System (CEMS) Engine Configuration</i>	15 Dec 2025
Technical Order (TO) 00-85-20	<i>Engine Shipping Instructions</i>	1 May 2025
AWB-330A	<i>Propulsion System Type Certification</i>	17 Apr 2018
DD Form 250	<i>Material Inspection and Receiving Report</i>	1 Aug 2000

DD Form 1574	<i>Serviceable Tag – Material</i>	1 Oct 1988
DD Form 2794	<i>Cost and Software Data Reporting Plan</i>	16 May 2024
OTHER		
NIST SP 800-171 Rev. 3	<i>Protecting Controlled Unclassified Information in Nonfederal Systems and Organizations</i>	14 May 2024
Joint Aeronautical Commanders' Group (JACG)	<i>Aviation Critical Safety Item Management Handbook</i>	16 Mar 2011
AF6500 SAE	<i>Manufacturing Management Program</i>	12 Nov 2014
ASTM D1655-25a	<i>Standard Specification for Aviation Turbine Fuels</i>	17 Dec 2025
Contractor Documents		

3.0 MODIFICATIONS, ENHANCEMENTS, and UPGRADE SUPPORT - F139 SPARE ENGINES

3.1 PW4062-3 Engine

The Contractor shall fabricate and/or procure parts, assemble, test, and deliver zero-time (new Life Limited Parts) F139 spare engine cores. The zero-time F139 spare engine cores shall be produced to the same standard as the engine cores produced for installation on KC-46A production aircraft. The Contractor shall produce and deliver all spare engine cores in accordance with P&W (OEM-managed) configuration at the time of award.

The Contractor shall deliver spare engine configuration data/records to the Government for each spare engine; the Government will be responsible for entry of spare engine data into the USAF G081 Mobility Aircraft Data Systems real-time database (*Core Automated Maintenance System for Mobility (CAMS-FM/G081)*).

CDRL A001, DI-MISC-80508B/T, Technical Report-Study/Services: F139 Spare Engines, Subtitle: G081 Importable Excel File

CDRL A002, DI-MISC-80508B/T, Technical Report-Study/Services: F139 Spare Engines, Subtitle: Vital Statistics Logbook (VSL)

3.2 Manufacturing Management Program

The Contractor shall maintain manufacturing management.

3.2.1 Bill of Material (BOM)

The Contractor Shall maintain a high-level detail (Core) BOM for the F139 spare engines associated with this procurement.

3.3 Contractor Furnished Fuel

The Contractor shall furnish petroleum/fuel necessary for use in the performance of this contract for engine assembly and acceptance testing. The Contractor shall comply with *American Society for Testing and Materials* (ASTM) D1655-21c, *Standard Specification for Aviation Turbine Fuels*, with the following additives: Inhibitor, Corrosion/Lubricity Improver (CI/LI) (MIL-PRF-25017), *Inhibitor, Icing, Fuel System, High Flash NATO Code Number S-1745* (FSII) (MIL-DTL-85470B), an approved antioxidant (AO) material listed in MIL-DTL-5624 or MIL-DTL-83133, an approved static dissipater additive (SDA) listed in MIL-DTL-5624 or MIL-DTL-83133, a thermal stability improver additive as described in MIL-DTL-83133.

3.4 Logistics

3.4.1 Comprehensive Engine Management System (CEMS)

The Contractor reporting responsibilities are described in Technical Order (TO) 00-25-254-1, *Comprehensive Engine Management System (CEMS) Engine Configuration*. The Contractor shall:

- Gather all CEMS reporting data items for each new spare engine, sub-part/components, etc., and deliver the data electronically to the Government (IAW CDRL A004, DI-MGMT-81324C/T) at the time of spare engine DD-250 Certification and Delivery (for Government entry of data into the CEMS database: via *CEMS Data Load File*). The Contractor will not be required to access/enter engine data (initial load) into the CEMS Central Database (CDB) for spare engines associated with this DO SOW.
- CEMS data provided by the Contractor shall include all hardware/components installed within each engine, to include, but not limited to: Configuration Item Identifier (CII), Work Unit Code, Item Nomenclature, Life Limited Part verification (Yes/No), Part Number, and Serial Number. The “CEMS Data Load File” delivered by the Contractor shall be compatible with CEMS/D042 formats (see tailored CDRL item below).

CDRL A004, DI-MGMT-81324C/T, Comprehensive Engine Management System (CEMS)
Reporting, CEMS Data Load File

3.4.2 Engine Item Unique Identification (IUID) Markings

The Contractor shall mark the engine data plate with IUID in accordance with MIL-STD-130N, *Identification Marking of U.S. Military Property*, or equivalent standard. Any currently IUID marked engine components shall be maintained for this program. The Contractor shall upload the IUID assigned numbers and information into the Government IUID Registry following the Using Service’s guidance and procedures. The IUID shall contain, at a minimum: applicable Enterprise Identifier (EID), Serial Number, and Part or Identifying Number (PIN). IUID markings shall be visible during normal handling, be readable, withstand the specified environment and cleaning processes, and be as permanent as the service life of the unit or at least to the unit’s point of rebuild.

All engines shall be delivered with their EECs marked on their Pb Sensor Inspection Plate with the “born-on” information of the sensors IAW Service Bulletin PW4ENG 73-221 for calendar time tracking per SB EEC131-73-62.

3.4.3 Packaging, Handling, Storage, and Transportation (PHS&T)

The Contractor shall be responsible for the preservation and packaging of all necessary components and engines for shipment IAW the Packaging Instruction specified on Air Force Materiel Command (AFMC) Form 158, *Packaging Requirements*. The Contractor shall follow packaging, handling, storage, transportation practices in accordance with applicable existing Technical Orders (TOs) and/or in accordance with MIL-STD-2073-1E, *Standard Practice for Military Packaging*. The Contractor shall place a special tag, DD Form 1574, *Serviceable Tag - Material*, inside the package along with the assembly as well as outside the package. In addition, all marking, and labeling of unit and shipping containers shall be in accordance with MIL-STD-129R, *Military Marking for Shipment and Storage*.

3.4.4 Shipping of Spare Engines

The Contractor shall prepare the preserved/wrapped spare engines for shipping, including preparing spare engines within protection-coverings (e.g. reusable protection bags) appropriate for mode(s) of shipping in accordance with MIL STD -2073-1E, *Standard Practice for Military Packaging*. The Contractor shall be responsible for all shipment/transportation costs of USAF spare engines to designated CONUS locations. The contractor shall transport engines IAW TO 00-85-20 (Engine Shipping Instructions), AFMCI24-201 (Packaging, Handling, Storage, and Transportation Acquisition and Sustainment Product Support Instruction), and Standards Practices Manual (Part Number 585005) Section 70-80-00 following all guidelines, AFI's, Technical Instructions, and other references mentioned therein.

4.0 TEST AND EVALUATION - PRODUCT ACCEPTANCE

The Contractor shall perform the level of test and evaluation required to ensure all spare engines associated with this procurement, will perform in their intended environment in support of the KC-46A weapon system. The Contractor shall adhere to FAA and military certifications, and Operational Safety, Suitability & Effectiveness (OSS&E) baselines.

4.1 Acceptance Test Plans and Acceptance Test Procedures (ATPs)

The Contractor shall develop/update and maintain the Acceptance Test Plan for F139 spare engines. The test plan defines the acceptance test, performance objectives, criteria, and detailed procedures for the acceptance of all F139 spare engines and associated components and articles that will be accepted and delivered to the Government.

CDRL B005, DI-QCIC-80553A/T, Acceptance Test Plan, Subtitle: F139 Spare Engines

5.0 SYSTEMS ENGINEERING/PROJECT MANAGEMENT (SEPM)

The Contractor shall perform any systems engineering and project management efforts applicable to the spare engines production.

6.0 PROGRAM MANAGEMENT

6.1 Contract Work Breakdown Structure (CWBS)

The Contractor shall develop, deliver, and maintain a CWBS and a CWBS Dictionary in accordance with MIL-STD-881F, *Work Breakdown Structures for Defense Material Items*. The Contractor shall request and receive Government approval before making any changes to the CWBS and/or associated definitions impacting reporting level Work Breakdown Structure elements.

CDRL A006, DI-MGMT-81334D/T, Contractor Work Breakdown Structure

6.2 Integrated Master Schedule (IMS)

The Contractor shall develop, deliver, implement, and maintain an IMS that identifies the contracted F139 Spare Engine buy scope and external critical dependencies, such as P&W engine production and deliveries, test/verification activities, DD-250 activities, and spare engine shipment-related information/timelines (examples only - not all inclusive). The Contractor shall map the schedule information to the CWBS.

CDRL A007, DI-MGMT-81861C/T, Integrated Program Management Data and Analysis Report, Integrated Master Schedule (IMS)

6.3 Cost and Software Data Reporting (CSDR)

The Contractor shall systematically collect and electronically report to the Cost Assessment Data Enterprise the actual contract costs and technical information based on the Office of the Secretary of Defense (OSD) Deputy Director Cost Assessment (DDCA) approved CSDR Plans in accordance with the Department of Defense Manual (DoDM) 5000.04, *Cost and Software Data Reporting (CSRD) Manual*.

The Contractor shall present its approach to satisfy the CSDR guidelines contained in DoDM 5000.04, *Cost and Software Data Reporting (CSDR) Manual*, and the requirements in the OSD DDCA-approved CSDR Plan for the contract, DD Form 2794, *Cost and Software Data Reporting Plan*, and the related Resource Distribution Table.

CDRL A008, DI-FNCL-81765B/T, Contractor Business Data Report (DD Form 1921-3), Contractor Cost Data Reporting (CCDR)

CDRL A009, DI-FNCL-82162, Cost and Hour Report (Flex File), Contractor Cost Data Reporting (CCDR)

CDRL A010, DI-MGMT-82165, Technical Data Report

CDRL A011, DI-MGMT-82164, Quantity Data Report, Contractor Cost Data Reporting (CCDR)

6.4 Risk Management

The Contractor shall execute a Risk Management Plan for this procurement.

6.5 Government Industry Data Exchange Program (GIDEP)

The Contractor shall participate in the GIDEP. The Contractor shall generate “alerts” when applicable, acquire “alerts” from GIDEP Operations Center (Naval Surface Warfare Center, Corona Division, Corona, CA 92878-8000), and take corrective action when “alerts” impact the spare engine production program.

CDRL A016, DI-QCIC-80125B/T, Government Industry Data Exchange Program (GIDEP) Alert/Safe-Alert Report

CDRL A017, DI-QCIC-80126B/T, Government Industry Data Exchange Program (GIDEP) Alert Response

CDRL A018, DI-QCIC-80127A/T, Government Industry Data Exchange Program (GIDEP) Annual Progress Report

7.0 CONFIGURATION MANAGEMENT

7.1 Contract Change Proposal (CCP)

The Contractor shall propose non-technical contract changes using existing Government CCP processes. Additionally, the Contractor shall notify designated AFLCMC/LPK-1 Procurement Contracting Officer (PCO).

CDRL A012, DI-ADMN-81401C/T, Contract Change Proposals (CCPs)

7.2 Specification Change Notice (SCN)

All specification changes generated under this contract shall be subject to Government approval and submitted as an SCN. Additionally, the Contractor shall provide any SCN to the designated AFLCMC/LPK-1 Procurement Contracting Officer (PCO).

CDRL A013, DI-SESS-80643E/T, Specification Change Notice (SCN)

7.3 Deviations and Waivers

The Contractor shall prepare requests for deviation and waivers using existing Government processes. All deviation and waiver requests shall also be provided to the designated AFLCMC/LPK-1 Procurement Contracting Officer (PCO).

CDRL A014, DI-SESS-80640E/T, Request for Variance (RFV)

CDRL A015, DI-SES-81732/T, Request for Waiver (RFW)

8.0 DATA MANAGEMENT

The Contractor shall implement data management procedures for this procurement.

CDRL B019, DI-MGMT-81453B, Data Accession List (DAL)

9.0 RESERVED

10.0 RESERVED

11.0 RESERVED

There will be no GFP/GFE under this procurement for spare engines.

12.0 RESERVED

13.0 QUALITY ASSURANCE

The Contractor or its supplier(s) shall maintain a quality system that addresses the elements of ISO 9001- ANSI/ASQC Q9001, quality system model for Quality Assurance in Design and Development, Production, Installation, and Servicing.

13.1 Foreign Object Damage (FOD) Prevention Program

The Contractor shall maintain existing FOD Prevention Programs. The Contractor shall assure FOD Prevention is accomplished, and work areas maintained in a manner to prevent foreign objects from entering and remaining in product structure(s). The FOD Prevention Industry Guideline shall be used as a guide.

14.0 CONTINUED AIRWORTHINESS AND REGULATORY COMPLIANCE

The contractor will continue to maintain all current airworthiness and regulatory compliance associated with the KC-46A/F139 engine.

15.0 SYSTEM SAFETY

16.0 ENVIRONMENTAL FACTORS/ENVIRONMENT SAFETY AND OCCUPATIONAL HEALTH (ESOH)

17.0 SECURITY

18.0 KC-46 INFORMATION TECHNOLOGY

19.0 RESERVED